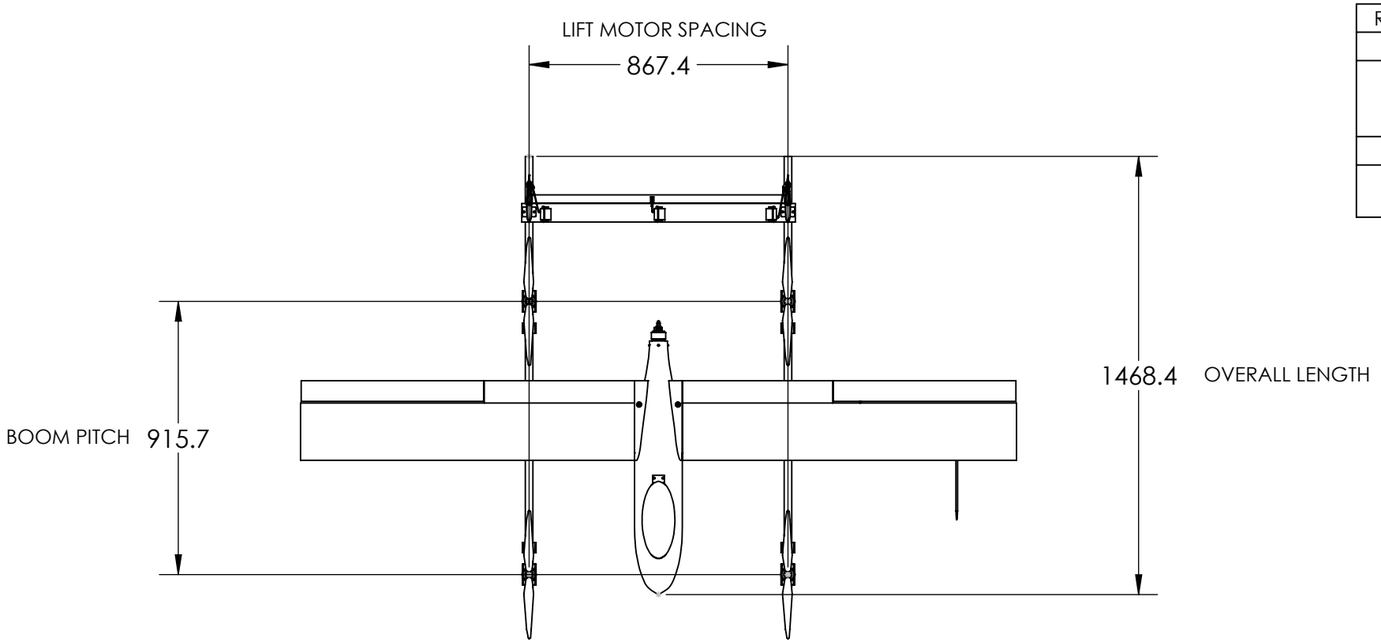
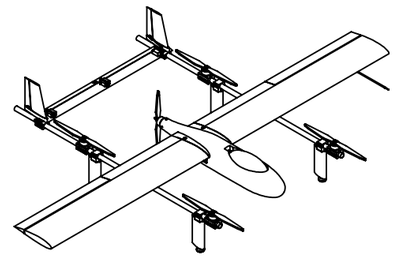
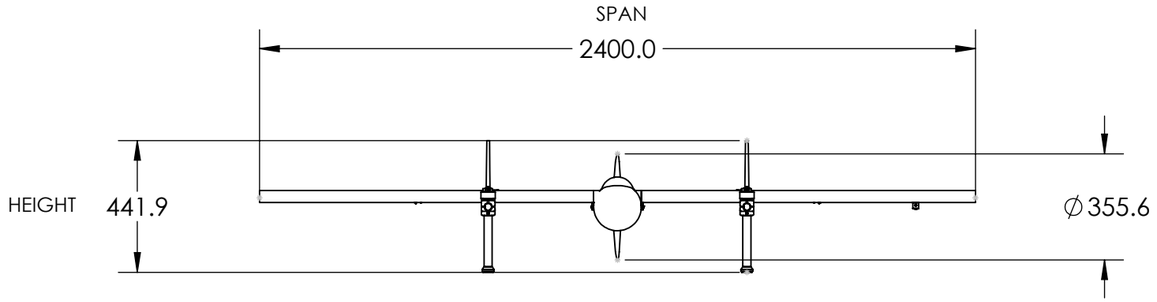
REV	DATE	DESCRIPTION	BY
A	2026-10-18	CONCEPT GEOMETRY FOR DESIGN REVIEW (v0.3.1), 5 SHEETS	DBF
B	2026-10-21	HARDWARE SPEC APPLIED: SPLIT MOUNTS AND CLAMPS, BOOM SPLICE, M5 WING BOLTS, RUDDERS, ALL FASTENERS, TOLERANCES, FASTENER SCHEDULE; 9 SHEETS	DBF
C	2026-10-23	DRAWING LAYOUT AND BALLOONING CORRECTED	DBF
D	2026-10-25	WING RESIZED FOR R11 STALL (CHORD 266.6, S 0.640 M2); SIZES FROM PARAMS	DBF



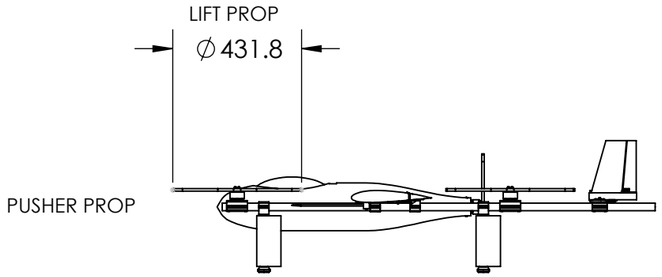
PLAN 1:20



ISOMETRIC 1:30



FRONT ELEVATION 1:20



SIDE ELEVATION 1:20

NOTES
1. DIMENSIONS IN MM, THIRD-ANGLE PROJECTION. PROPS SHOWN PARKED FORE-AFT (NOTIONAL: THE ESCs CANNOT PARK THEM).
2. SIZED CONCEPT, MTOW 6.45 kg, SPAN 2400 (analysis/results.json). EVERY FASTENER IS MODELLED: SHEETS 2, 3 AND 8.
3. TRANSPORT BREAKDOWN: 2 x M5 WING BOLTS, 8 x M3 BOOM-CLAMP SCREWS, 4 x M3 TAIL-SHOE SCREWS; THE JOINER SLIDES OUT OF THE SADDLE (SHEET 4).
4. NO-PAYLOAD FLIGHT: 0.5 kg BALLAST IN THE PAYLOAD BAY KEEPS THE CG AT 28 % MAC. PUSHER PROP GROUND CLEARANCE 42 mm.

GENERAL TOLERANCES	UNLESS NOTED
PRINTED PARTS (DESKTOP FDM)	±0.5 % OF SIZE, MIN ±0.5
CONTROLLED BORES, INSERT HOLES	FIT TABLE FT02 / FT03 / FT08 (hardware.json)
HOLE POSITION	Ø0.4; MOTOR PATTERN Ø0.2
CUT TUBES (ISO 2768-m), ANGLES	±0.5 (120–400), ±0.8 (400–1000); ±1°

DESIGN-BUILD-FLY
UC BERKELEY CAPSTONE
REV D 2026-10-25
FIRST-ARTICLE M1–M7 APPLY

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN MM
THIRD-ANGLE PROJECTION
TOLERANCES: SEE TABLE
FITS: hardware.json
INTERPRET GEOMETRIC TOLERANCING PER:
MATERIAL SEE PARTS LIST
FINISH SEE PARTS LIST
DO NOT SCALE DRAWING
NEXT ASSY | USED ON | APPLICATION

	NAME	DATE
DRAWN	DBF	2026-10-25
CHECKED	—	
ENG APPR.		
MFG APPR.		
Q.A.		

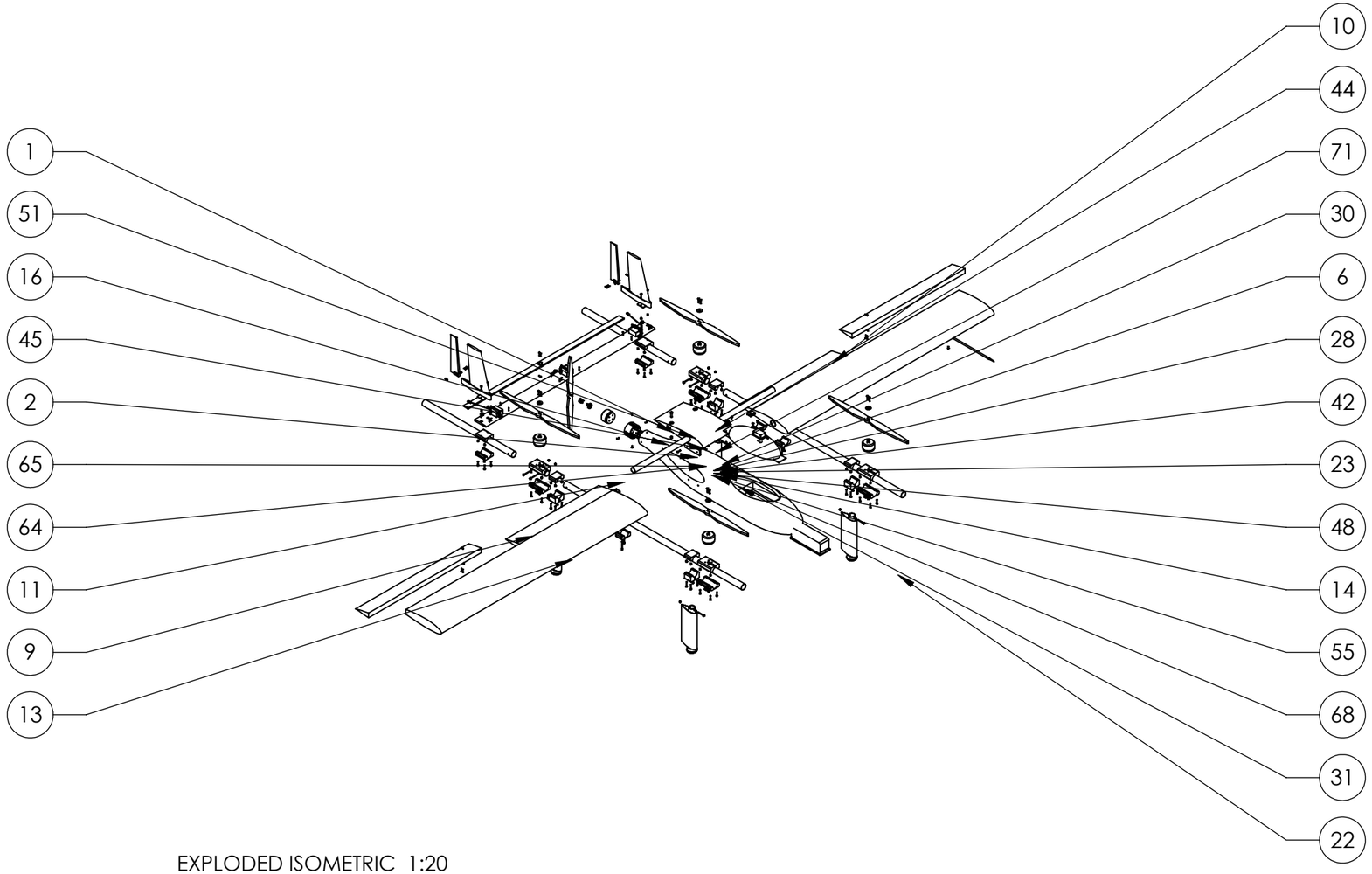
COMMENTS:

DESIGN-BUILD-FLY

TITLE:
LIFT + CRUISE VTOL
GENERAL ARRANGEMENT

SIZE B | DWG. NO. DBF-DRW-001 | REV D
SCALE: 1:20 | WEIGHT: | SHEET 1 OF 9

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EXPLODED ISOMETRIC 1:20

NOTES
1. EXPLODED LAYOUT (VTOL_Exploded.SLDASM): PARTS MOVED BY THE OFFSETS IN cad/params.mjs; FASTENERS PULLED 50 mm ALONG THEIR AXES.
2. ONE BALLOON PER MAJOR PART TYPE; ITEM NUMBERS ARE THOSE OF THE PARTS LIST (SHEET 3). FASTENERS, INSERTS AND SMALL PRINTED HARDPOINTS ARE NOT BALLOONED: SEE SHEETS 3, 8 AND 9.
3. JOINT IDs (J01–J20) AND FITS (FT01–FT11): docs/research/hardware-spec.md.

GENERAL TOLERANCES	UNLESS NOTED
PRINTED PARTS (DESKTOP FDM)	±0.5 % OF SIZE, MIN ±0.5
CONTROLLED BORES, INSERT HOLES	FIT TABLE FT02 / FT03 / FT08 (hardware.json)
HOLE POSITION	Ø0.4; MOTOR PATTERN Ø0.2
CUT TUBES (ISO 2768-m), ANGLES	±0.5 (120–400), ±0.8 (400–1000); ±1°

DESIGN-BUILD-FLY
UC BERKELEY CAPSTONE
REV D 2026-10-25
FIRST-ARTICLE M1–M7 APPLY

		UNLESS OTHERWISE SPECIFIED:		NAME	DATE
		DIMENSIONS IN MM THIRD-ANGLE PROJECTION TOLERANCES: SEE TABLE FITS: hardware.json	DRAWN	DBF	2026-10-25
			CHECKED	—	
			ENG APPR.		
			MFG APPR.		
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.		
		MATERIAL SEE PARTS LIST	COMMENTS:		
NEXT ASSY	USED ON	FINISH SEE PARTS LIST			
APPLICATION		DO NOT SCALE DRAWING			

DESIGN-BUILD-FLY

TITLE:
EXPLODED ASSEMBLY WITH BALLOONS

SIZE	DWG. NO.	REV
B	DBF-DRW-001	D

SCALE: 1:10 | WEIGHT: | SHEET 2 OF 9

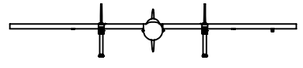
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ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Fuselage_Pod	Fuselage pod 0.85 x 0.16 m, foam +…	1
2	Canopy	Avionics hatch blister (EST)	1
3	F07_M3x8_BHCS	ISO 7380-1 button head M3x0.5 x 8 A2…	11
4	I01_M3_Insert	ruthex RX-M3x5.7 heat-set insert OD…	67
5	Battery_Tray	Printed PETG battery tray 165 x 56 x 3…	1
6	Battery_6S_LiPo	6S 6500 mAh LiPo envelope (dims EST)…	1
7	FC_Tray	Printed PETG flight-controller tray 60…	1
8	Flight_Controller	CoreWing F405 Wing V2 envelope 52 x 32…	1
9	Wing_Panel_R	Right wing panel, NACA 2415 (EST), XPS…	1
10	Wing_Panel_L	Left wing panel, NACA 2415 (EST), XPS…	1
11	Aileron_R	Right aileron, 26 % chord, 55 % of…	1
12	Aileron_L	Left aileron, 26 % chord, 55 % of…	1
13	Spar_Tube	6061-T6 1.000 x 0.058 in spar socket…	2
14	Wing_Joiner	6061-T6 0.875 x 0.058 in joiner, 760 mm	1
15	Joiner_Stop	Printed PETG joiner stop 22.4 x 10…	2
16	Wing_Saddle	Wing centre-section saddle	1
17	Bolt_Pillar	Printed PETG wing-bolt pillar dia 16 x…	2
18	Root_Tongue	Printed PETG panel root tongue 20 x…	2
19	F09_M5x20_BHCS	ISO 7380-1 button head M5x0.8 x 20 A2…	2
20	N02_M5_Nylock	DIN 985 nylon-insert lock nut M5x0.8…	2
21	F08_M3x12_SHCS	ISO 4762 M3x0.5 x 12 A2 (18-8)…	4
22	Boom_Front	Front boom, 6061-T6 1.000 x 0.035 in…	2
23	Boom_Rear_Stub	Rear boom stub carrying the tail…	2
24	Splice_Plug	Printed PETG splice plug 23.2 x 80…	2
25	F04_M3x40_SHCS	ISO 4762 M3x0.5 x 40 A2 (18-8)…	8

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
26	W01_M3_Washer	DIN 125 (vendor) M3 flat washer OD 7…	28
27	N01_M3_Nylock	DIN 985 nylon-insert lock nut M3x0.5…	8
28	Motor_Mount_Upper	Printed PA6-CF motor mount, upper half…	4
29	Motor_Mount_Cap	Printed PA6-CF motor mount, lower cap…	4
30	Lift_Motor	SunnySky X4110S lift motor (envelope…	4
31	Lift_Prop	JeeFly 1755 carbon lift prop (planform…	4
32	Prop_Plate	X4110S prop holding plate dia 20 x…	4
33	F02_M3x16_SHCS	ISO 4762 M3x0.5 x 16 A2 (18-8)…	36
34	F01_M3x8_SHCS	ISO 4762 M3x0.5 x 8 A2 (18-8) stainless	20
35	Boom_Clamp_F_Upper	Printed PETG front boom clamp, upper…	2
36	Boom_Clamp_F_Lower	Printed PETG front boom clamp, lower…	2
37	Boom_Clamp_R_Upper	Printed PETG rear boom clamp, upper…	2
38	Boom_Clamp_R_Lower	Printed PETG rear boom clamp, lower…	2
39	Spar_Collar	Printed PETG spar collar 32 x 46…	2
40	Rear_Hardpoint	Printed PETG rear wing hardpoint 60 x…	2
41	F06_M3x35_SHCS	ISO 4762 M3x0.5 x 35 A2 (18-8)…	4
42	Gear_Clamp_Upper	Printed PETG gear clamp, upper half…	4
43	Gear_Clamp_Lower	Printed PETG gear clamp, lower half…	4
44	Landing_Leg	Gear leg: 6061 1.000 x 0.058 in offcut…	4
45	Gear_Fairing	Printed PETG landing-leg fairing	4
46	F03_M3x20_SHCS	ISO 4762 M3x0.5 x 20 A2 (18-8)…	16
47	F05_M3x30_SHCS	ISO 4762 M3x0.5 x 30 A2 (18-8)…	4
48	Tail_Clamp_Upper	Printed PETG tail clamp, upper half…	2
49	Tail_Clamp_Cap	Printed PETG tail clamp, lower cap…	2
50	Fin_Shoe	Printed PETG fin root shoe	2

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
51	Fin	Vertical fin, NACA 0009, lofted taper…	2
52	Rudder	Rudder, 30 % of the local fin chord (2…	2
53	Crush_Sleeve	Printed PETG crush sleeve 7 x 3.4 x…	4
54	Servo_Tray	Printed PETG servo tray (L-bracket)	5
55	Servo	EMAX ES3005HV servo 38.6 x 18.8 x 34.9…	5
56	F10_M2x10_SHCS	ISO 4762 M2x0.4 x 10 A2 (18-8)…	20
57	N03_M2_Hex_Nut	DIN 934 hex nut M2x0.4, s 4 h 1.6	30
58	Control_Horn	Nylon control horn 11 x 11 x 20 from…	5
59	Horn_Backplate	Kit horn backplate 20 x 11 x 1.5 (EST)…	3
60	F11_M2x14_PAN	— M2x0.4 x 14 —: Control-horn screws…	10
61	Pushrod_Rudder	Du-Bro 230 2-56 pushrod cut to 100 mm…	2
62	Horn_Hardpoint	Printed PETG aileron horn hardpoint 20…	2
63	Pushrod_Aileron	Du-Bro 230 2-56 pushrod cut to 45 mm…	2
64	H_Tail	Horizontal stabiliser, NACA 0009…	1
65	Elevator	Elevator, 30 % chord, NACA 0009, tape…	1
66	Pushrod_Elevator	Du-Bro 230 2-56 pushrod cut to 33 mm…	1
67	Pusher_Cup	Printed PA6-CF pusher motor cup dia…	1
68	Pusher_Motor	SunnySky X4110S pusher motor, same…	1
69	Pusher_Adapter	X4110S bolt-on prop adapter, supplied	1
70	F13_M3x8_SHCS	— M3x0.5 x 8 — (supplied)	4
71	Pusher_Prop_14in	APC 14 x 8.5EP pusher prop (planform…	1
72	Prop_Washer	Pusher prop washer dia 17.8 x 2.5, dia…	1
73	N04_M6_Bullet_Nut	M6 x 1.0 bullet prop nut, supplied…	1
74	Pitot_Tube	Pitot probe under the left wing…	1
75	Pitot_Clamp	Printed PETG pitot clamp 20 x 20 x…	1

NOTES
1. ITEM NUMBERS ARE THE BALLOONS OF SHEET 2; QTY IS PER AIRCRAFT. FASTENERS, NUTS, WASHERS AND INSERTS ARE ONE LINE PER TYPE.
2. PART NUMBER = SOLIDWORKS FILE (cad/model). FASTENER IDs F01–F13, N01–N04, W01, I01 ARE THOSE OF cad/hardware.json AND SHEET 8. FULL DESCRIPTIONS: PART FILE PROPERTIES.
3. PURCHASED ITEMS, PRICES AND VENDORS: docs/purchasing/C0_BOM.csv.
4. MATERIALS: 6061-T6 TUBES; A2 STAINLESS FASTENERS; BRASS INSERTS; PETG PRINTS EXCEPT MOTOR MOUNTS AND PUSHER CUP IN PA6-CF; XPS FOAM.



REFERENCE: FRONT ELEVATION 1:50

GENERAL TOLERANCES	UNLESS NOTED
PRINTED PARTS (DESKTOP FDM)	±0.5 % OF SIZE, MIN ±0.5
CONTROLLED BORES, INSERT HOLES	FIT TABLE FT02 / FT03 / FT08 (hardware.json)
HOLE POSITION	Ø0.4; MOTOR PATTERN Ø0.2
CUT TUBES (ISO 2768-m), ANGLES	±0.5 (120–400), ±0.8 (400–1000); ±1°

DESIGN-BUILD-FLY
UC BERKELEY CAPSTONE
REV D 2026-10-25
FIRST-ARTICLE M1–M7 APPLY

		UNLESS OTHERWISE SPECIFIED:		NAME	DATE
		DIMENSIONS IN MM THIRD-ANGLE PROJECTION TOLERANCES: SEE TABLE FITS: hardware.json	DRAWN	DBF	2026-10-25
			CHECKED	—	
			ENG APPR.		
			MFG APPR.		
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.		
		MATERIAL SEE PARTS LIST	COMMENTS:		
NEXT ASSY	USED ON	FINISH SEE PARTS LIST			
APPLICATION		DO NOT SCALE DRAWING			

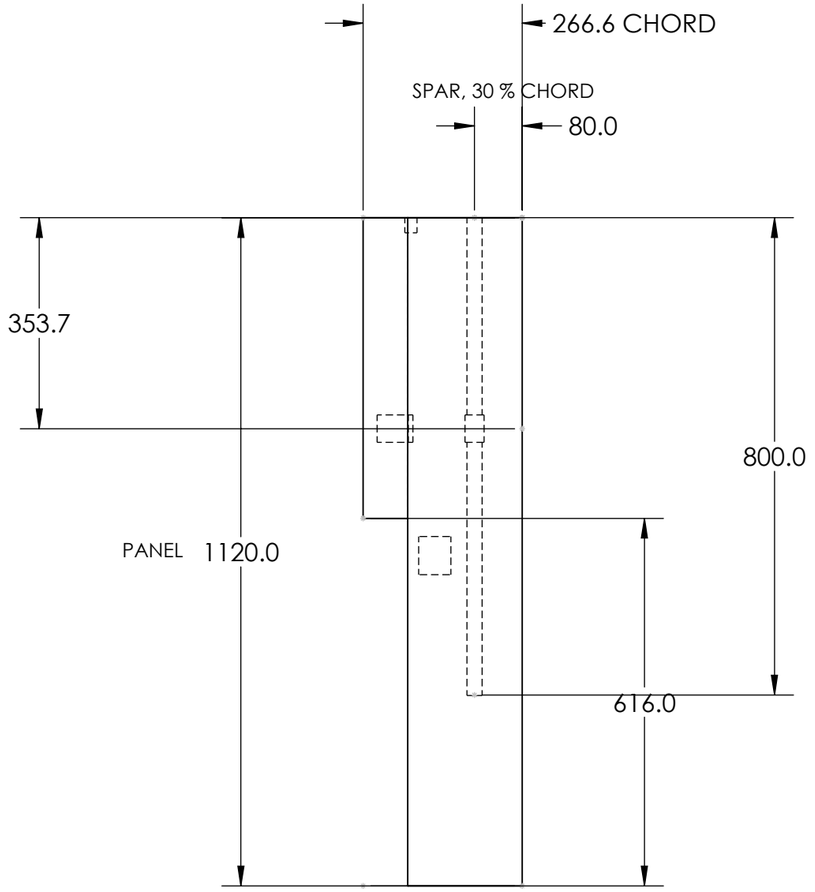
DESIGN-BUILD-FLY

TITLE:
LIFT + CRUISE VTOL
PARTS LIST

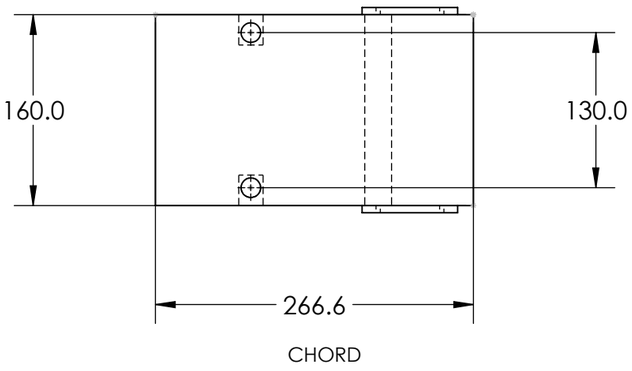
SIZE	DWG. NO.	REV
B	DBF-DRW-001	D

SCALE: 1:50 | WEIGHT: | SHEET 3 OF 9

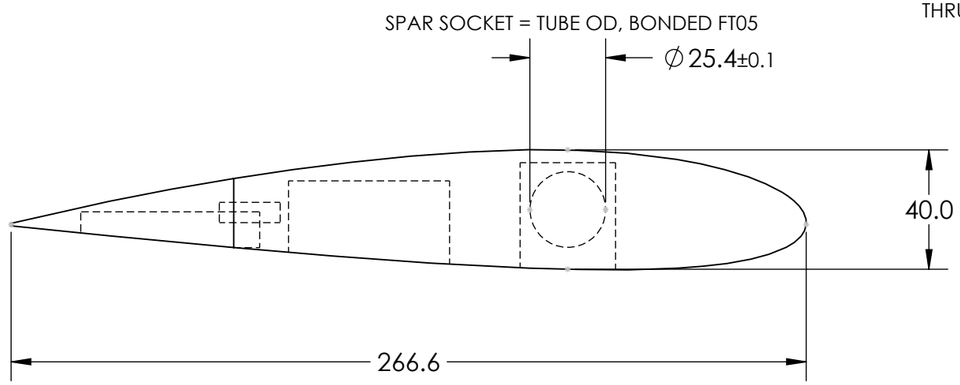
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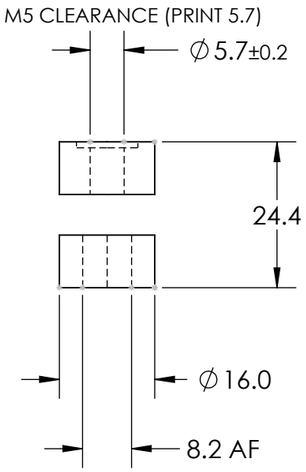
WING PANEL, RIGHT (LEFT IS MIRROR) 1:10



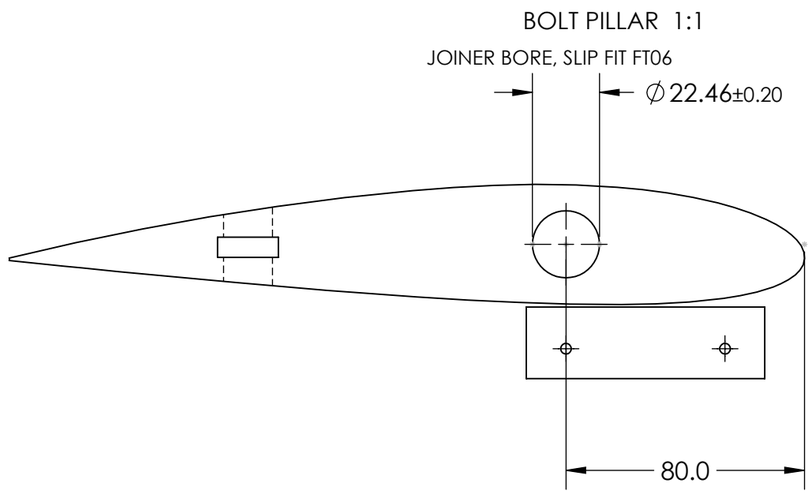
WING SADDLE 1:5



ROOT SECTION 1:2



BOLT PILLAR 1:1



JOINER AT 30 % CHORD

SADDLE SECTION 1:2

NOTES
1. NACA 2415, XPS CORE, PACKING-TAPE SKIN. AILERON 26 % CHORD, 2 % HINGE GAP, OUTER 55 % OF SPAN (616 DIM). SPAR SOCKET 800 DIM, BOOM STATION 353.7 FROM THE ROOT, ROOT THICKNESS 40.0 AT 30 % CHORD. SADDLE: 2X BOLT PILLARS ∅16 IN ∅16.4 BORES AT 130 (DIM), 70 % CHORD.
2. SPAR SOCKET 1.000 x 0.058 in 6061-T6 BONDED 800 mm INTO THE ROOT AT 30 % CHORD (FT05, FOAM-SAFE GLUE).
3. JOINER 0.875 x 0.058 in, 760 mm: SLIP FIT IN THE SOCKETS (FT01, -0.08/+0.53 DIAMETRAL, MEASURE M1) AND THE SADDLE (FT06); NOT BONDED, CAPTURED BY TWO PRINTED STOPS 300 mm INSIDE THE SOCKETS.
4. WING BOLT J10: F09 ISO 7380 M5 x 20 THROUGH THE PILLAR AND THE ROOT TONGUE (SHEET 9) INTO N02 M5 NYLOCK, 1.0 N·m EST; HEAD IN A ∅10.3 x 1.0 SPOT-FACE; NUT POCKET 8.2 AF x 9.0 FROM THE LOWER SURFACE (SCREW END 0.5 PAST THE NUT).
5. PANEL POCKETS (SPAR COLLAR, REAR HARDPOINT, SERVO TRAY, TONGUE, PITOT CLAMP) CUT TO THE PART OUTLINES OF SHEET 9, BONDED WITH FOAM-SAFE GLUE.
6. SADDLE TABS: 2 x F08 M3 x 12 PER SIDE INTO RX-M3x5.7 INSERTS IN THE POD FRAME RAILS (J11).

GENERAL TOLERANCES	UNLESS NOTED
PRINTED PARTS (DESKTOP FDM)	±0.5 % OF SIZE, MIN ±0.5
CONTROLLED BORES, INSERT HOLES	FIT TABLE FT02 / FT03 / FT08 (hardware.json)
HOLE POSITION	Ø0.4; MOTOR PATTERN Ø0.2
CUT TUBES (ISO 2768-m), ANGLES	±0.5 (120–400), ±0.8 (400–1000); ±1°

DESIGN-BUILD-FLY
UC BERKELEY CAPSTONE
REV D 2026-10-25
FIRST-ARTICLE M1–M7 APPLY

		UNLESS OTHERWISE SPECIFIED:		NAME	DATE
		DIMENSIONS IN MM THIRD-ANGLE PROJECTION TOLERANCES: SEE TABLE FITS: hardware.json	DRAWN	DBF	2026-10-25
			CHECKED	—	
			ENG APPR.		
			MFG APPR.		
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.		
		MATERIAL SEE PARTS LIST	COMMENTS:		
NEXT ASSY	USED ON	FINISH SEE PARTS LIST			
APPLICATION		DO NOT SCALE DRAWING			

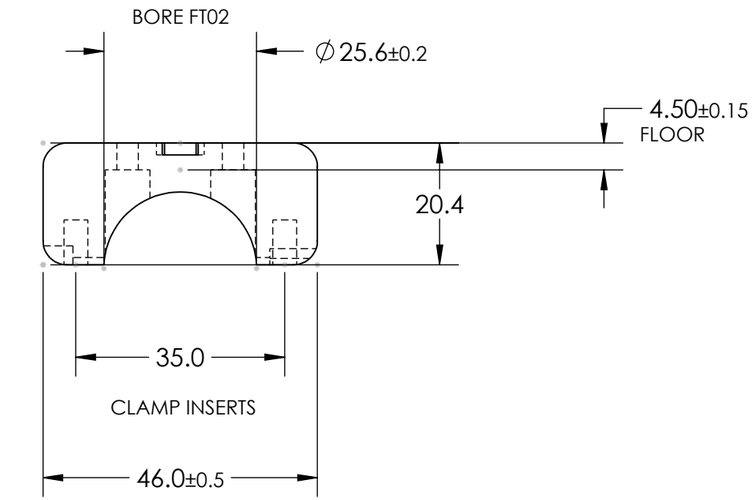
DESIGN-BUILD-FLY

TITLE:
WIGN PANEL, SADDLE AND WING BOLT

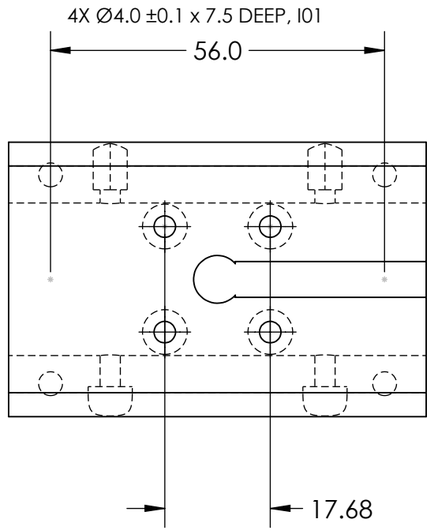
SIZE	DWG. NO.	REV
B	DBF-DRW-001	D

SCALE: 1:10 WEIGHT: SHEET 4 OF 9

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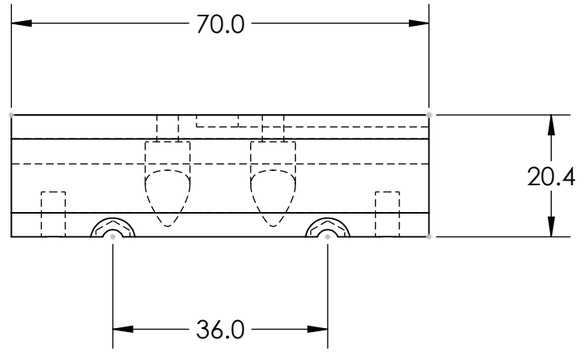


UPPER HALF, SECTION 1:1



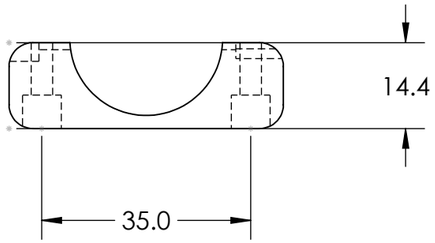
4X Ø3.4 THRU ON Ø25 B.C., POS Ø0.2

UPPER HALF, TOP 1:1

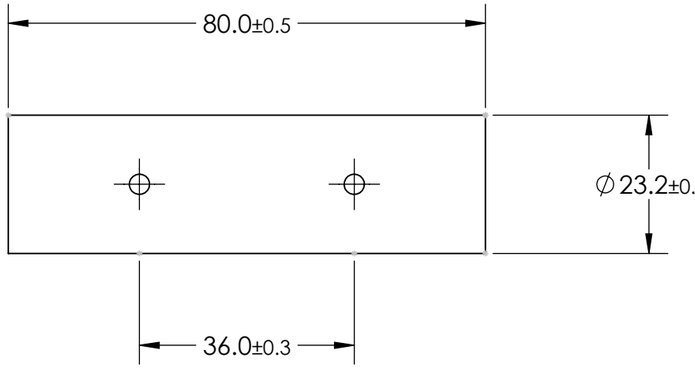


2X Ø3.4 THRU Z, SPLICE J04

UPPER HALF, SIDE 1:1

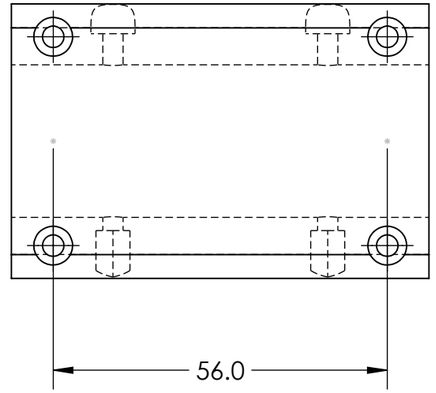


CAP, SECTION 1:1



2X Ø3.4 THRU

SPLICE PLUG 1:1



4X Ø3.4 THRU, CBORE Ø6.5 x 5.6

CAP, BOTTOM 1:1

NOTES
1. PA6-CF (HDT 173 °C), 4 UPPER HALVES + 4 CAPS. BORES PRINTED 0.2 UNDER (COUPON M2), REAMED TO 25.6 ±0.2 (FT02). SPLIT GAP 1.2 (0.6 EACH HALF).
2. J01 MOTOR: 4X F01 M3 x 8 + W01 FROM INSIDE THE BORE BEFORE THE HALVES CLOSE (CBORE ∅7.5 FROM THE BORE SIDE, HOLES DRILLED 3.4 AFTER PRINTING); ENGAGEMENT 3.0, LIMIT 3.3 (M4); 0.6 N·m EST, LOCTITE 242.
3. J02 CLAMP: 4X F02 M3 x 16 FROM BELOW INTO THE INSERTS, 0.5 N·m EST, NO THREADLOCKER, WITNESS MARK.
4. J04 SPLICE (REAR STATION): BOOMS BUTT AT THE MOUNT CENTRE OVER THE PLUG; 2X F04 M3 x 40 + W01 + N01 ALONG Z AT x ±18, HEAD CBORE ∅7.5 x 5.0 (+Z), NUT POCKET 5.7 AF x 8.0 (-Z); DRILL BOTH TUBE WALLS ∅3.4 THROUGH THE MOUNT, ±0.3.
5. WIRE GROOVE 6 x 2 ON THE TOP FACE TOWARDS +X; CENTRE RELIEF ∅8 x 2 FOR THE MOTOR BOSS.

GENERAL TOLERANCES	UNLESS NOTED
PRINTED PARTS (DESKTOP FDM)	±0.5 % OF SIZE, MIN ±0.5
CONTROLLED BORES, INSERT HOLES	FIT TABLE FT02 / FT03 / FT08 (hardware.json)
HOLE POSITION	Ø0.4; MOTOR PATTERN Ø0.2
CUT TUBES (ISO 2768-m), ANGLES	±0.5 (120–400), ±0.8 (400–1000); ±1°

DESIGN-BUILD-FLY
UC BERKELEY CAPSTONE
REV D 2026-10-25
FIRST-ARTICLE M1–M7 APPLY

		UNLESS OTHERWISE SPECIFIED:
		DIMENSIONS IN MM THIRD-ANGLE PROJECTION TOLERANCES: SEE TABLE FITS: hardware.json
		INTERPRET GEOMETRIC TOLERANCING PER:
		MATERIAL SEE PARTS LIST
NEXT ASSY	USED ON	FINISH SEE PARTS LIST
APPLICATION		DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	DBF	2026-10-25
CHECKED	—	
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS:		

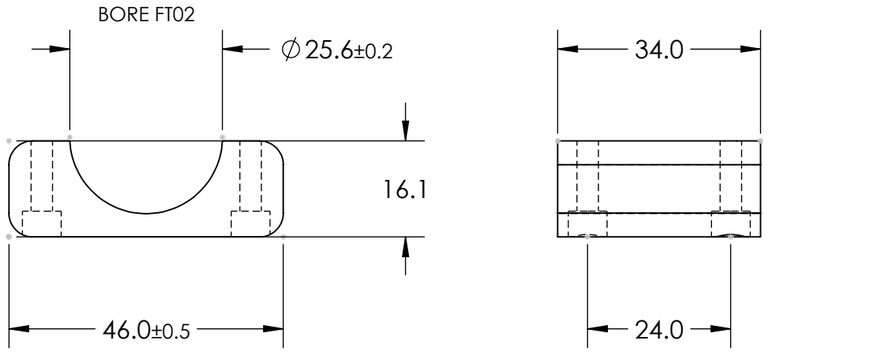
DESIGN-BUILD-FLY

TITLE:
MOTOR MOUNT
AND BOOM SPLICE

SIZE	DWG. NO.	REV
B	DBF-DRW-001	D

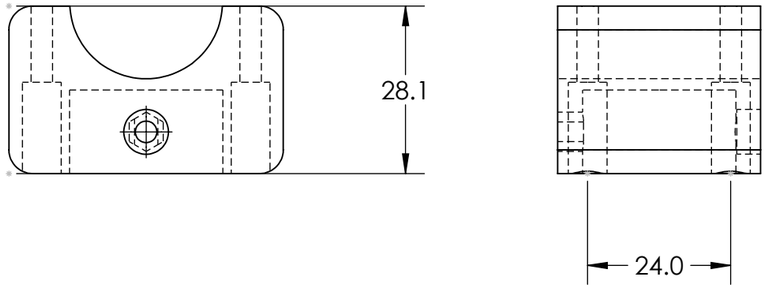
SCALE: 1:1 | WEIGHT: | SHEET 5 OF 9

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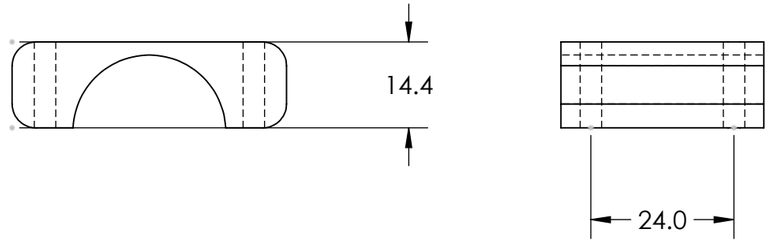


2X Ø3.4 THRU, CBORE Ø6.5 x 4.3 (FRONT) / x 7.5 (REAR)

FRONT / REAR BOOM CLAMP, LOWER (PETG) QTY 2 + 2 1:1

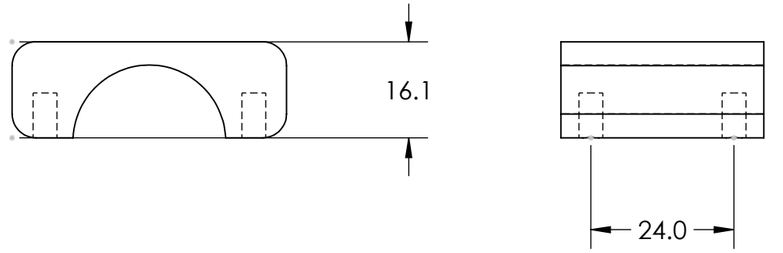


4X Ø3.4, CBORE Ø6.5 x 15.3



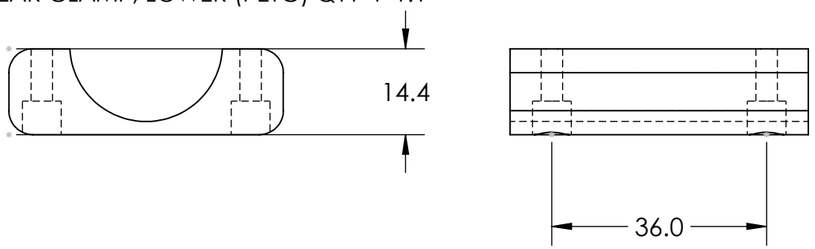
2X Ø3.4 THRU, DIAGONAL

FRONT BOOM CLAMP, UPPER (PETG) QTY 2 1:1



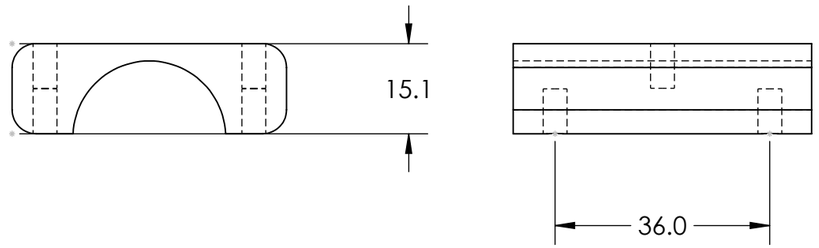
4X Ø4.0 ±0.1 x 7.5 DEEP, I01

GEAR CLAMP, LOWER (PETG) QTY 4 1:1



4X Ø3.4, CBORE Ø6.5 x 5.6

GEAR CLAMP, UPPER (PETG) QTY 4 1:1



4X Ø4.0 x 7.5 DEEP (I01), 2 FROM TOP

TAIL CLAMP, CAP (PETG) QTY 2 1:1

TAIL CLAMP, UPPER (PETG) QTY 2 1:1

NOTES
1. EACH PAIR: SECTION THROUGH THE BORE (LEFT), SIDE ALONG THE BOOM (RIGHT). ALL CLAMPS 46 WIDE, BORE ∅25.6 ±0.2 (FT02) PRINTED 0.2 UNDER AND REAMED, SPLIT GAP 1.2; BOOM AND GEAR CLAMPS 34 LONG, TAIL CLAMPS 50.
2. REAR BOOM CLAMP UPPER = FRONT UPPER, TOP AT 24.2; REAR LOWER = FRONT LOWER WITH THE DEEPER COUNTERBORE.
3. SCREWS, TORQUES, THREADLOCKER: SHEET 8 (J05 F06 / F04 ON A DIAGONAL INTO THE WING INSERTS, FIELD BREAKDOWN: J06 F03; J08 F02). LEG SOCKET AND J07 CROSS-BOLT: SHEET 7.

GENERAL TOLERANCES	UNLESS NOTED
PRINTED PARTS (DESKTOP FDM)	±0.5 % OF SIZE, MIN ±0.5
CONTROLLED BORES, INSERT HOLES	FIT TABLE FT02 / FT03 / FT08 (hardware.json)
HOLE POSITION	Ø0.4; MOTOR PATTERN Ø0.2
CUT TUBES (ISO 2768-m), ANGLES	±0.5 (120–400), ±0.8 (400–1000); ±1°

DESIGN-BUILD-FLY
UC BERKELEY CAPSTONE
REV D 2026-10-25
FIRST-ARTICLE M1–M7 APPLY

		UNLESS OTHERWISE SPECIFIED:		NAME	DATE
		DIMENSIONS IN MM THIRD-ANGLE PROJECTION TOLERANCES: SEE TABLE FITS: hardware.json	DRAWN	DBF	2026-10-25
			CHECKED	—	
			ENG APPR.		
			MFG APPR.		
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.		
		MATERIAL SEE PARTS LIST	COMMENTS:		
NEXT ASSY	USED ON	FINISH SEE PARTS LIST			
APPLICATION		DO NOT SCALE DRAWING			

DESIGN-BUILD-FLY

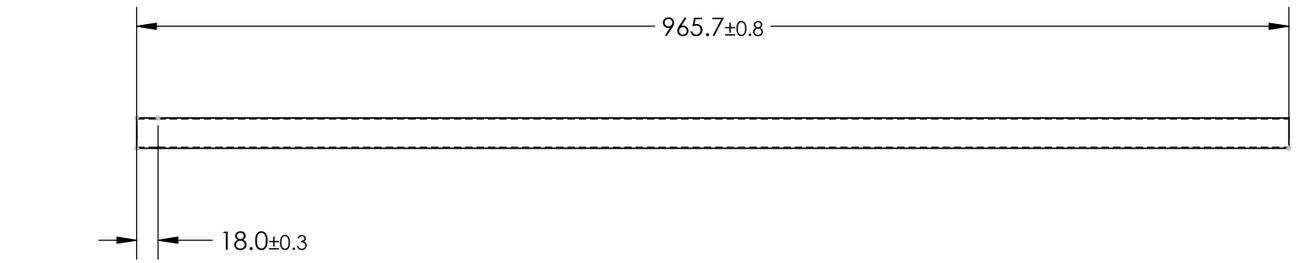
TITLE:
BOOM, GEAR AND TAIL CLAMPS

SIZE	DWG. NO.	REV
B	DBF-DRW-001	D

SCALE: 1:1 | WEIGHT: | SHEET 6 OF 9

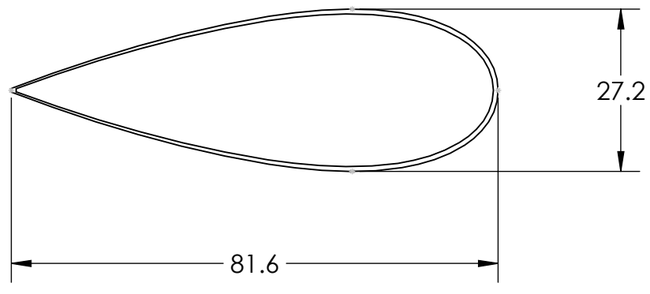
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ITEM	PART	STOCK (6061-T6)	CUT LENGTH mm	TOL mm (ISO 2768-m)	QTY	HOLES
T01	Wing spar socket	1.000 x 0.058 in	800.0	±0.8	2	none; bonded in the panel (FT05)
T02	Wing joiner	0.875 x 0.058 in	760.0	±0.8	1	none; slip fit FT01/FT06, lap if tight
T03	Front boom	1.000 x 0.035 in	965.7	±0.8	2	1 x Ø3.4 cross hole (Z) 18 ±0.3 from the splice end, drilled through the mount
T04	Rear boom stub	1.000 x 0.035 in	485.9	±0.8	2	1 x Ø3.4 cross hole (Z) 18 ±0.3 from the splice end
T05	Landing leg	1.000 x 0.058 in	205.0	±0.5	4	1 x Ø3.4 cross hole (X) 7 ±0.3 from the top end, drilled through the gear clamp



Ø3.4 THRU (Z), SPLICE BOLT

FRONT BOOM, SPLICE END AT LEFT 1:5



LEG BORE 25.6

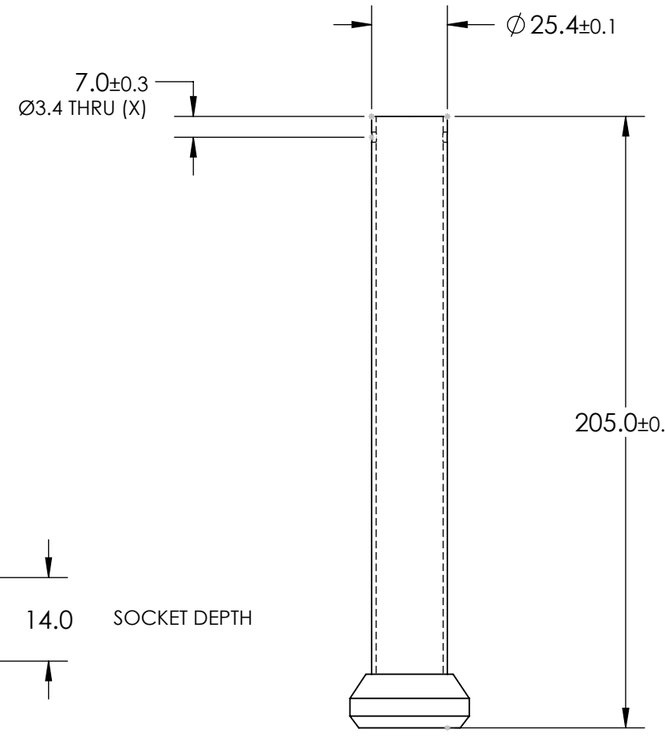
CHORD

LEG FAIRING SECTION (PETG, 170 LONG) QTY 4 1:1

SOCKET DEPTH

BLIND SOCKET FT03

LEG SOCKET IN THE GEAR CLAMP LOWER (SHEET 6) 1:1



LANDING LEG 1:2

NOTES
1. LEGS ARE CUT FROM THE SPAR-SOCKET TUBE OFFCUT: 2 x 800 + 4 x 205 = 2420 mm OF A 96 in (2438 mm) LENGTH. BOOMS: 2 x (965.7 + 485.9) = 2903 mm FROM TWO 72 in LENGTHS.
2. END SQUARENESS 0.5 (EST). DEBURR ALL CUT ENDS AND HOLES. TUBE OD ±0.10, WALL ±0.05 (VENDOR, HS §5a): MEASURE ON RECEIPT (M1).
3. BOOMS BUTT AT THE REAR MOTOR MOUNT CENTRE OVER THE PRINTED SPLICE PLUG (SHEET 5). LEG: J07 F05 M3 x 30 + W01 + N01 CROSS-BOLT, HEAD CBORE Ø7.5 x 4.0, NUT POCKET 5.7 AF x 4.3; LEG DRILLED Ø3.4 THROUGH THE CLAMP.
4. LEG FAIRINGS: 3:1 TEARDROP (NACA 0033) SLIPPED OVER EACH LEG, TRAILING EDGE AFT, VASE-MODE 0.8 mm WALL (EST), CLOCKED AGAINST THE GEAR CLAMP AND TAPED; NO FASTENERS (DRAG: analysis/aero/aero_results.json).

GENERAL TOLERANCES	UNLESS NOTED
PRINTED PARTS (DESKTOP FDM)	±0.5 % OF SIZE, MIN ±0.5
CONTROLLED BORES, INSERT HOLES	FIT TABLE FT02 / FT03 / FT08 (hardware.json)
HOLE POSITION	Ø0.4; MOTOR PATTERN Ø0.2
CUT TUBES (ISO 2768-m), ANGLES	±0.5 (120–400), ±0.8 (400–1000); ±1°

DESIGN-BUILD-FLY
UC BERKELEY CAPSTONE
REV D 2026-10-25
FIRST-ARTICLE M1–M7 APPLY

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN MM
THIRD-ANGLE PROJECTION
TOLERANCES: SEE TABLE
FITS: hardware.json
INTERPRET GEOMETRIC TOLERANCING PER:
MATERIAL: SEE PARTS LIST
FINISH: SEE PARTS LIST
NEXT ASSY | USED ON
APPLICATION | DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	DBF	2026-10-25
CHECKED	—	
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS:		

DESIGN-BUILD-FLY

TITLE:
TUBE CUT LIST
AND LANDING LEG

SIZE B | DWG. NO. DBF-DRW-001 | REV D
SCALE: 1:2 | WEIGHT: | SHEET 7 OF 9

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ID	STANDARD	SIZE	QTY	ORDER	JOINTS	TORQUE N·m	THREADLOCKER	VENDOR / PART NO.
F01	ISO 4762 (DIN 912)	M3x0.5 x 8	20	30	J01, J12	0.6	Loctite 242, one drop on the last 2-3 threads only; keep off the PA6-CF (TDS	Bolt Depot 6379
F02	ISO 4762 (DIN 912)	M3x0.5 x 16	36	50	J02, J08, J09, J03	0.5	none (brass insert in plastic); witness mark	Bolt Depot 6382
F03	ISO 4762 (DIN 912)	M3x0.5 x 20	16	24	J06	0.5	none	Bolt Depot 6383
F04	ISO 4762 (DIN 912)	M3x0.5 x 40	8	12	J04, J05b	0.5	none (nylock)	Bolt Depot 6387
F05	ISO 4762 (DIN 912)	M3x0.5 x 30	4	8	J07	0.5	none (nylock)	Bolt Depot 6385 (M3 x 30 socket cap; number and price not verified, EST)
F06	ISO 4762 (DIN 912)	M3x0.5 x 35	4	8	J05a	0.5	none	Bolt Depot 6386
F07	ISO 7380-1 button head (vendor lists no standard	M3x0.5 x 8	11	20	J12, J13, J16, J15	0.4	none	Bolt Depot 7219
F08	ISO 4762 (DIN 912)	M3x0.5 x 12	4	12	J11	0.5	none	Bolt Depot 6381
F09	ISO 7380-1 button head (vendor lists no standard	M5x0.8 x 20	2	6	J10	1	none (nylock)	Bolt Depot 6635
F10	ISO 4762 (DIN 912)	M2x0.4 x 10	20	30	J14	0.15	none	Bolt Depot 6365
F11	— (supplier states M2x14 only)	M2x0.4 x 14	10	0	J17	0.15	none (nylon horn)	Amazon (kit item 7) B0B58ZSC8J
F12	ISO 4762 (DIN 912)	M3x0.5 x 16	8	0	J03	0.8	Loctite 242 on the screw tip (metal into motor hub)	Bolt Depot (from F02 or F08 stock) 6382 / 6381
F13	— (supplied; ISO 4762 geometry assumed in the CAD)	M3x0.5 x 8	4	0	J03b	0.8	Loctite 242 (metal to metal)	SunnySky (in motor box) —
I01	ruthex RX-M3x5.7	M3 x 5.7	67	100	J02 J03b J05 J06 J08 J09 J11 J12 J13 J15 J16	—	—	Amazon (ruthex) B08BCRZZS3
N01	DIN 985 nylon-insert lock nut (vendor	M3x0.5 x 4	8	16	J04, J07	—	—	Bolt Depot 4792
N02	DIN 985 nylon-insert lock nut	M5x0.8 x 5	2	6	J10	—	—	Bolt Depot 4794
N03	DIN 934 hex nut	M2x0.4 x 1.6	30	100	J14, J17	—	—	Bolt Depot 7362
N04	M6x1.0 bullet (acorn) prop nut, supplied with motor	M6x1.0 x 19.5	1	0	J03b	—	—	SunnySky (in motor box) —
W01	DIN 125 (vendor)	M3 7 x 0.5	28	100	J01, J12, J04, J07	—	—	Bolt Depot 4513

NOTES
(docs/research/hardware-spec.md §2d, §4)
1. QTY PER AIRCRAFT; ORDER INCLUDES SPARES. F12 PROP SCREWS AND F13 ADAPTER SCREWS: LENGTH PROVISIONAL, CHECK M5 (HUB THICKNESS NOT PUBLISHED).
2. TORQUES MARKED EST IN THE SPEC: CONFIRM INSERT TORQUE-OUT >= 2x ON A COUPON (M3). NO TORQUE DRIVER ON HAND: HANG A KNOWN MASS FROM THE HEX KEY, T = m g L (0.6 N·m = 1.02 kg AT L 60 mm; 0.5 N·m = 0.85 kg; 1.0 N·m = 1.70 kg).
3. LOCTITE 242 ONLY ON METAL-TO-METAL THREADS (F01 MOTOR BASE, F12/F13 HUB SCREWS), ONE DROP ON THE LAST 2–3 THREADS, KEPT OFF THE PA6-CF. NEVER ON NYLOCKS.
4. SCREWS IN BRASS INSERTS GET NO THREADLOCKER: TORQUE, THEN A PAINT WITNESS MARK CHECKED PRE-FLIGHT. REPLACE A NYLOCK THAT TURNS BY HAND.
5. INSERTS RX-M3x5.7 IN Ø4.0 ±0.1 HOLES >= 7.0 DEEP (FT08, INTERFERENCE INTENDED); SOLDERING IRON ~20 °C ABOVE THE PRINT TEMPERATURE, SET PER MATERIAL.
6. CLEARANCE HOLES: ISO 273 NORMAL (M2 2.4, M3 3.4, M5 5.5) PRINTED AT +0.2 AND CHECKED ON THE HOLE COUPON (M2); MOTOR-PATTERN HOLES DRILLED TO 3.4 AFTER PRINTING.
7. MOTOR SCREWS: TIP <= 3.2 mm PROUD OF THE FLOOR BEFORE EVERY MOTOR INSTALL (M4); 3.3 mm REACHES THE WINDINGS.
8. FIELD BREAKDOWN (R4): 2 x F09, 8 x BOOM-CLAMP SCREWS (F06/F04), 4 x F02 SHOE SCREWS; 14 SCREWS IN A LABELLED BAG.

GENERAL TOLERANCES	UNLESS NOTED
PRINTED PARTS (DESKTOP FDM)	±0.5 % OF SIZE, MIN ±0.5
CONTROLLED BORES, INSERT HOLES	FIT TABLE FT02 / FT03 / FT08 (hardware.json)
HOLE POSITION	Ø0.4; MOTOR PATTERN Ø0.2
CUT TUBES (ISO 2768-m), ANGLES	±0.5 (120–400), ±0.8 (400–1000); ±1°

DESIGN-BUILD-FLY
UC BERKELEY CAPSTONE
REV D 2026-10-25
FIRST-ARTICLE M1–M7 APPLY

NEXT ASSY	USED ON	
APPLICATION		

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN MM
THIRD-ANGLE PROJECTION
TOLERANCES: SEE TABLE
FITS: hardware.json
INTERPRET GEOMETRIC TOLERANCING PER:
MATERIAL: SEE PARTS LIST
FINISH: SEE PARTS LIST
DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	DBF	2026-10-25
CHECKED	—	
ENG APPR.		
MFG APPR.		
Q.A.		

COMMENTS:

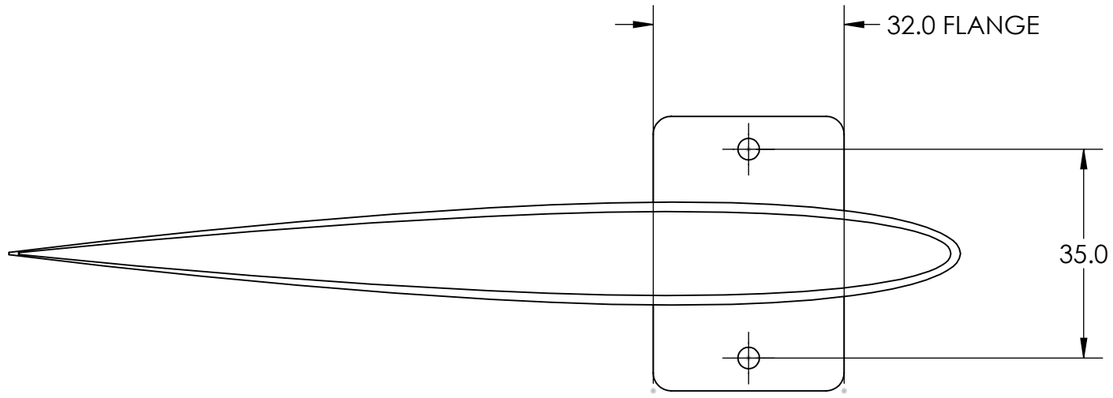
DESIGN-BUILD-FLY

TITLE: FASTENER SCHEDULE AND TORQUES

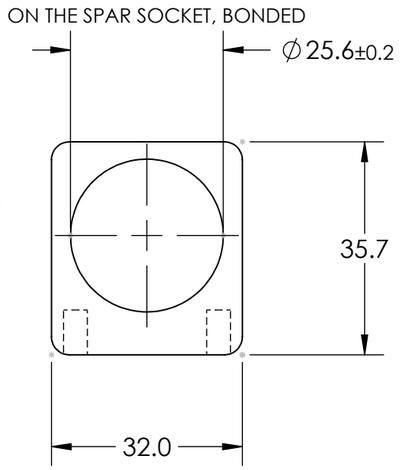
SIZE	DWG. NO.	REV
B	DBF-DRW-001	D

SCALE: 1:1 | WEIGHT: | SHEET 8 OF 9

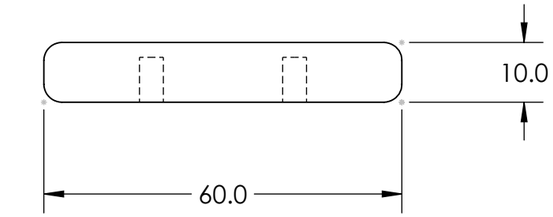
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2X Ø3.4 THRU

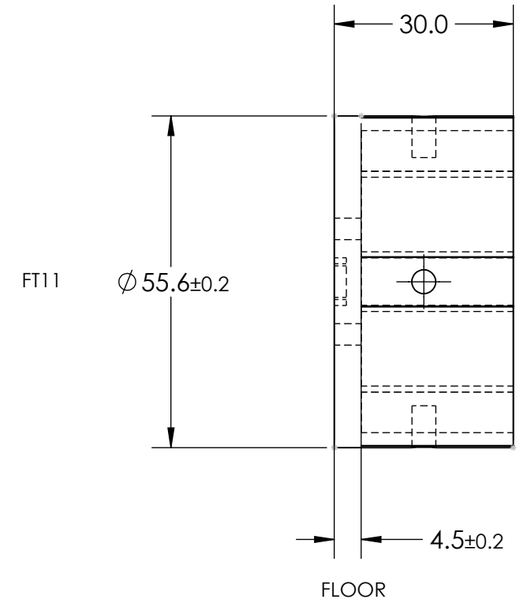


SPAR COLLAR (PETG) QTY 2 1:1

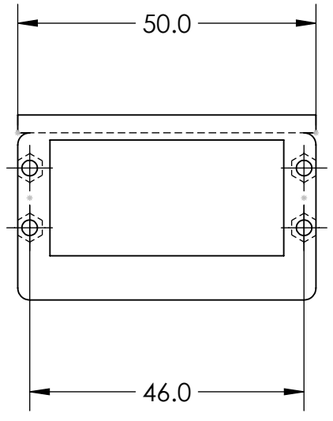


REAR HARDPOINT (PETG) QTY 2 1:1

FIN SHOE (PETG) QTY 2 1:1

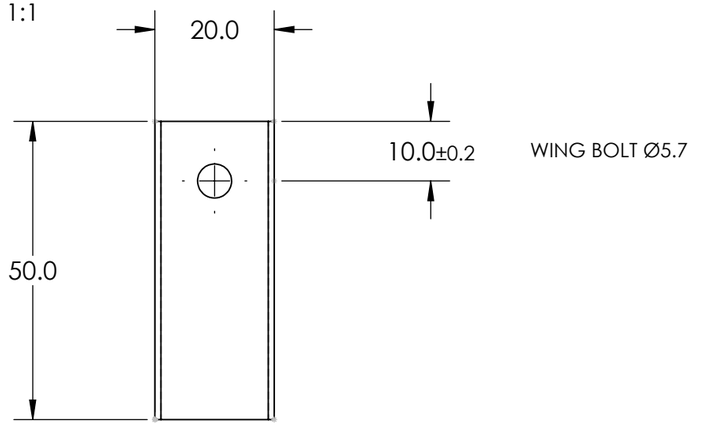


PUSHER CUP (PA6-CF) QTY 1 1:1



4X Ø2.6 + M2 NUT POCKET 4.2 AF x 2

SERVO TRAY (PETG) QTY 5 1:1



ROOT TONGUE (PETG) QTY 2 1:1

NOTES
1. FIN SHOE J09: FLOOR 3.0, WALLS 20 x 1.6 AROUND THE FIN ROOT (FIN BONDED IN); 2X F02 M3 x 16 THROUGH THE FLANGE AND A CRUSH SLEEVE 7 x 3.4 x 6.8 INTO THE TAIL-CLAMP INSERTS.
2. SPAR COLLAR J05a, REAR HARDPOINT J05b: BONDED FLUSH INTO THE WING LOWER SURFACE AT THE BOOM STATION, INSERTS ∅4.0 x 7.5 FACE DOWN.
3. PUSHER CUP J12: 4X F01 + W01 FROM INSIDE INTO THE MOTOR BASE, 4X F07 RADIAL THROUGH THE POD RING. SERVO TRAY J14: 4X F10 + N03, TAB HOLES PROVISIONAL (M6).
4. ROOT TONGUE J10: 25 mm BONDED IN THE PANEL ROOT, 25 mm IN THE SADDLE PILLAR. OTHER PRINTED PARTS: DIMENSIONS IN THE PART FILES (PARTS LIST). TORQUES: SHEET 8.

GENERAL TOLERANCES	UNLESS NOTED
PRINTED PARTS (DESKTOP FDM)	±0.5 % OF SIZE, MIN ±0.5
CONTROLLED BORES, INSERT HOLES	FIT TABLE FT02 / FT03 / FT08 (hardware.json)
HOLE POSITION	Ø0.4; MOTOR PATTERN Ø0.2
CUT TUBES (ISO 2768-m), ANGLES	±0.5 (120–400), ±0.8 (400–1000); ±1°

DESIGN-BUILD-FLY
UC BERKELEY CAPSTONE
REV D 2026-10-25
FIRST-ARTICLE M1–M7 APPLY

		UNLESS OTHERWISE SPECIFIED:		NAME	DATE
		DIMENSIONS IN MM THIRD-ANGLE PROJECTION TOLERANCES: SEE TABLE FITS: hardware.json	DRAWN	DBF	2026-10-25
			CHECKED	—	
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		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.		
		MATERIAL SEE PARTS LIST	COMMENTS:		
NEXT ASSY	USED ON	FINISH SEE PARTS LIST			
APPLICATION		DO NOT SCALE DRAWING			

DESIGN-BUILD-FLY

TITLE:
PRINTED PARTS
HARDPOINTS AND TRAYS

SIZE	DWG. NO.	REV
B	DBF-DRW-001	D

SCALE: 1:1 | WEIGHT: | SHEET 9 OF 9

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